



Effects of buffer gas composition on autoignition



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ABSTRACT

This work quantifies the chemical kinetic and thermal effects of buffer gas composition on autoignition of three fuels at conditions relevant to engines, combustors, and experimental facilities used to study ignition kinetics. Computational simulations of autoignition of *iso*-octane, *n*-heptane, and of *n*-butanol were used to characterize the effects of buffer gas composition on ignition delay time and heat release rate. Stoichiometric mixtures, $\phi = 1.0$, and a temperature range of 600–1100 K were considered. *iso*-octane and *n*-heptane were studied at initial pressures of 9.0 atm and 60.0 atm, and *n*-butanol was studied at initial pressures of 3.2 atm and 60.0 atm. Two dilution levels of buffer gas to O₂ of 3.76:1 (mole basis) and 5.64:1 were considered ($\sim 21\%$ and $\sim 15\%$ O₂ respectively, mole basis). The fuels and simulation conditions were selected based on the relevance to engine operating conditions and previously published ignition studies. The buffer gases considered were argon, nitrogen, water, and carbon dioxide. Simulation results predicted changes of greater than a factor of 2 in ignition delay time and heat release rate as a function of buffer gas composition in the negative temperature coefficient (NTC) region for *n*-heptane and *iso*-octane. Outside the NTC region, the predicted effects of changes in buffer gas composition were small ($<20\%$); however, experimental data for *n*-heptane indicate larger effects of buffer gas composition on ignition delay time at higher temperatures (>2 factor of 2). The heat release rates were also sensitive to buffer gas composition, with carbon dioxide exhibiting relatively high levels of early and late heat release relative to the other buffer gases. Sensitivity analysis of the third-body collision efficiencies for the buffer gases showed the effects of uncertainties in the third body collision efficiencies on ignition delay time and heat release rate. The results highlight the significance of buffer gas composition on low-temperature combustion chemistry, particularly via H₂O₂ and HO₂ decomposition and recombination reactions.

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1. Introduction

Dilution strategies, such as exhaust gas recirculation (EGR), are important tools to achieve high efficiency, low pollutant emissions combustion. There are multiple mechanisms by which EGR can improve combustion performance, including direct cooling or heating, dilution, and potential chemical kinetic interactions through three-body reactions and through trace reactive components in the EGR gases, to name a few. The chemical kinetic and thermal effects of EGR are important as they play significant roles on reaction rates and thereby affect autoignition times and heat release rates. Moreover, thermal, dilution, and chemical kinetic effects are often convolved. For example, thermal effects of buffer gas composition include changes in the specific heat capacity of the fuel/air mixture which affect compression heating and heat transfer losses, and thereby impact chemical reaction rates. Dilution impacts reaction

rates (chemical kinetic effects) and heat transfer rates (thermal effects). Chemical kinetic effects also include the impact of third-body collision efficiencies. It is challenging to isolate the effects of EGR composition in internal combustion engine (ICE) studies due to the complexities of the combustion systems and the often limited access for engine diagnostics. Despite these challenges, there have been valuable experimental and computational ICE studies that have investigated some of the thermal and chemical kinetic effects of buffer gases on autoignition, or combustion phasing, and exhaust gas emissions [1–5].

Ladommatos et al. [1] investigated EGR effects which they categorized as thermal (due to changes in the specific heat capacity of the mixture), dilution (due to changing the oxygen concentration of the mixture), and chemical kinetic effects (associated with dissociation of CO₂) on ignition delay times and exhaust emissions using CO₂ as an EGR substitute in a single cylinder, direct injection diesel engine. The authors found that the magnitudes of the three effects on ignition delay times and exhaust emissions ranked dilution effects as most important, chemical kinetic effects second, and thermal effects as the least important at the conditions studied.

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With increased EGR (i.e., CO₂) Ladommatos et al. [1] reported decreased NO_x emissions and increased ignition delay times, unburnt hydrocarbons, and particulate emissions. A single-cylinder engine operated in homogeneous charge compression ignition (HCCI) mode was used by Sjöberg et al. [2] to study EGR effects via simulated complete stoichiometric products (i.e., N₂, CO₂, and H₂O) and real EGR on autoignition behavior of gasoline and gasoline surrogates (including *iso*-octane and primary reference fuel, PRF, blends of *iso*-octane and *n*-heptane). The authors reported enhanced EGR dilution effects (due to changing the oxygen concentration of the mixture) in fuels with two-stage heat release compared to fuels with single stage heat release, stronger acceleration of autoignition by water (considered a chemical effect, where the authors stated that early reactions were enhanced by the presence of H₂O) for two-stage fuels relative to single stage fuels, and a higher sensitivity to the specific heat capacity of the mixture for single stage fuels. In the work by Machrafi et al. [3], the authors used simulated EGR gases during HCCI operation of a single cylinder engine. They considered the chemical kinetic effects of trace reactive species found in real EGR (the authors added small quantities of CO, NO, and CH₂O to the simulated EGR mixtures), dilution effects (specifically the oxygen concentration of the mixture), and the thermal effects of the specific heat capacity of the EGR gases. Machrafi et al. [3] proposed the formation and consumption rates of OH were critical to the effects of the EGR gases on autoignition. In the work by Al-Qurashi et al. [4], the authors spent considerable effort to isolate and quantify the thermal (specific heat capacity of the mixture), dilution (oxygen concentration of the mixture), and chemical kinetic (dissociation of CO₂) effects of CO₂ on the reactivity of soot produced in co-flow flames and compared the results with soot produced in diesel engines using CO₂ with simulated and actual EGR. The authors found the thermal effects were the most important (45%) on the soot oxidation rate compared to the dilution (35%) and chemical kinetic effects (20%). Sjöberg and Dec [5] studied EGR effects on autoignition of ethanol in an HCCI operated engine and compared results to their previous study of gasoline and PRF blends [2]. The authors found that ethanol was less sensitive to EGR effects. Sjöberg and Dec attributed the difference in sensitivity to the shorter chain length of ethanol and generally lower intermediate temperature heat release rates. The authors also noted that water decreased ignition delay times for ethanol, but less than was observed for gasoline and PRF blends.

In other devices, buffer gas composition is used to control state conditions in combustion chemistry studies using shock tube (ST) and rapid compression machine (RCM) facilities. These facilities, which provide important experimental knowledge of chemical kinetics, rely on buffer gases ranging from monatomics (typically Ar and He) and diatomics (typically N₂) to polyatomics (typically CO₂) to control shock wave characteristics and thermodynamic conditions. The composition of the buffer gases affects the specific heat capacity of the test gas mixtures, the three-body collision reactions, and the heat transfer rates of the test gas mixtures. Few studies have focused on these important and interrelated mechanisms; however, some researchers have identified concerns particularly with autoignition studies. Davidson and Hanson compared predictions for autoignition times for *iso*-octane using two reaction mechanisms and conditions of $\phi = 1.0$, 2% O₂, and $P = 2$ atm [6]. They considered argon and nitrogen as buffer gases and found the change in specific heat capacity significantly affected the autoignition time (~50%) for one reaction mechanism, but not the other, at 1400 K. At NUI-Galway, Würmel et al. [7] conducted an experimental and computational study of buffer gas composition (Ar, Xe, He, N₂) effects in an RCM for 2,3-dimethylpentane ($\phi = 1.0$, 15 atm, and 650–950 K) [7], and a computational study of *iso*-octane ($\phi = 1.0$, 2.0 atm) and methane ($\phi = 0.5$, 1.8 atm) each with Ar and N₂ buffer gases at time scales relevant

to shock tubes. Würmel et al. [8] also conducted a separate computational study of their RCM which focused on the effects of buffer gas composition as related to piston head design and heat transfer. Würmel et al. concluded that argon tends to increase autoignition times in their RCM [7,8] and decrease autoignition times in their ST [7], with different physical phenomena dominating the behavior over the wide time scales relevant to the RCM and ST facilities. Shen et al. [9] conducted experimental shock tube studies of *iso*-octane over a range of equivalence ratios, pressures, and temperatures with argon and nitrogen as buffer gases, and made comparisons with three chemical kinetic mechanisms. Shen et al. [9] determined that argon autoignition times were 20% shorter than autoignition times of nitrogen buffer gas experiments, primarily due to the lower specific heat capacity of argon, and that the mechanisms accurately captured the trend but generally failed to accurately predict autoignition times.

In summary, previous studies have highlighted the value of identifying conditions that are most and least sensitive to buffer gas composition and the mechanisms causing such sensitivity. Isolating, in as much as possible, the chemical kinetic and thermal effects of buffer gas composition on fuel ignition characteristics will help interpret previous results as well as guide future efforts to leverage buffer gas composition as a combustion design tool. The objective of this study was to quantify specific chemical kinetic and thermal effects of buffer gas composition using reaction mechanisms that have been well validated and are widely accepted for representing the autoignition chemistry of three important fuels. The computational simulations of autoignition of test gas mixtures were used to identify conditions where buffer gas composition has the highest impact on ignition delay times and heat release rates, and to quantify the effects of uncertainties in the species-specific collision efficiencies used in three-body reactions.

The computational methods were applied across a range of thermodynamic conditions for *iso*-octane, *n*-heptane, and *n*-butanol. The autoignition characteristics of each of the fuels have been previously studied by several research groups, providing a range of data for comparison, and the experiments provide the basis for the conditions simulated in this study. *Iso*-octane and *n*-heptane, as primary reference fuels, have been extensively studied using rapid compression machines (*i*-C₈H₁₈ [10–16], *n*-C₇H₁₆ [10,16–19]) and shock tubes (*i*-C₈H₁₈ [9,20–24], *n*-C₇H₁₆ [20,25–30]) encompassing a wide range of conditions. Studies of *n*-butanol have occurred with greater frequency over the last 5 years, and autoignition data are available from rapid compression machines [31,32] and shock tubes [33–39].

2. Computational methods

Computational simulations were carried out using the CHEMKIN suite of software (version 10113, x64) [40] and assuming a closed 0-D homogeneous batch reactor at adiabatic, constant volume conditions. Default values from CHEMKIN were used for the solver tolerances and solver time-steps. Detailed chemical kinetic mechanisms were used for each of the fuels, and extensive information on the mechanisms can be found in the literature for *iso*-octane [41], *n*-heptane [19], and *n*-butanol [34]. Brief summaries of the development and validation of each reaction mechanism used in this study are provided below. The reaction mechanisms were selected due to the maturity and extensive validation that has been previously completed on the reaction chemistry. No modifications to reaction rates were made to the mechanisms considered in this study. The mechanisms do not consider NO_x chemistry. Table 1 provides a brief summary of the characteristics of the reaction mechanisms, including the range of conditions for which the mechanisms have been developed and validated. The

Table 1
Summary of the reaction mechanisms used in this work. Detailed descriptions of the development and validation of the reaction mechanisms can be found in the references listed.

Fuel	# Species	# Reactions	Validation devices and conditions ^a	References
<i>i</i> -C ₈ H ₁₈	874	3796	RCM, ST, JSR, PFR ($\phi = 0.3$ –1.5, $P = 1$ –45 atm, $T = 550$ –1700 K)	[41,42]
<i>n</i> -C ₇ H ₁₆	1795	7245	RCM, ST, JSR, PFR ($\phi = 0.3$ –1.5, $P = 1$ –50 atm, $T = 550$ –1700 K)	[19,41,43,46]
<i>n</i> -C ₄ H ₉ OH	243	1446	ST ($\phi = 0.5$ –2.0, $P = 1$ –8 atm, $T = 1100$ –1800 K) JSR ($\phi = 0.5$ –2.0, $P = 10$ atm, $T = 750$ –1100 K)	[34,44,45]

^a RCM = rapid compression facility, ST = shock tube, JSR = jet stirred reactor, PFR = plug flow reactor.

chemical kinetic mechanisms used in this study are provided in the [Supplemental material](#).

Initial conditions were selected based on relevance to internal combustion engine operating conditions and existing experimental ST and RCM data, particularly the initial pressures. The simulations were conducted at stoichiometric fuel-to-oxygen equivalence ratios ($\phi = 1.0$) and over a minimum temperature range of 600–1100 K in 25 K increments for all fuels. Two dilution levels of 3.76:1 and 5.64:1 (buffer gas to O₂ ratios, mole basis) were considered in this study. The dilution levels correspond to air levels of oxygen (or ~21% O₂) and a more dilute mixture with ~15% O₂, mole basis. Argon, nitrogen, carbon dioxide, and water vapor were each evaluated as buffer gases in this study. Devices such as rapid compression machines and internal combustion engines often use mixtures of buffer gases (e.g., 50% argon/50% nitrogen, or 10% carbon dioxide/15% water/75% nitrogen). In this study, only pure buffer gases (e.g., 100% nitrogen) were considered to isolate and maximize the effects of a particular component on the autoignition behavior. [Table 2](#) provides the initial conditions and mixture compositions studied.

2.1. *n*-Heptane mechanism

Simulations of *n*-heptane autoignition in this study used the reaction mechanism from the *n*-heptane ignition and speciation study of Karwat et al. [19]. Their *n*-heptane mechanism was largely based on the most recent Lawrence Livermore National Laboratory (LLNL) *n*-heptane chemical kinetic mechanism by Mehl et al. [41], which originates from the work of Curran et al. [43]. The *n*-heptane mechanism from Karwat et al. [19] can be categorized as having 25 distinct reaction classes which describe the low and high temperature chemical kinetic pathways for normal alkanes up to heptane, and a detailed small hydrocarbon (H₂ and C₁–C₄) submechanism updated from the work of Aul et al. [46]. Reaction classes are schemes that allow modelers to assign reaction rates to molecules that have not been studied either theoretically or experimentally based on similar structure. Karwat et al. [19] also modified the low-temperature chemistry (e.g., reactions involving alkylperoxy radicals, RO₂, and hydroperoxyalkyls, QOOH) based on theoretical calculations by Villano et al. [47,48] to improve agreement between model predictions and experimental measurements of heptane intermediates. Karwat et al. [19] found their mechanism

produced satisfactory results for their experimental results, in addition to agreeing with the validation targets tested from the Mehl et al. [41] and Curran et al. [43] studies.

2.2. *iso*-Octane mechanism

Autoignition simulations of *iso*-octane in this study were conducted using the LLNL *iso*-octane chemical kinetic mechanism (Version 3) [41] available online. Version 3 is the most recent available update of the LLNL *iso*-octane mechanism by Curran et al. [42]. Two sets of reaction blocks compose the *iso*-octane mechanism, the first is a chemistry set for hydrocarbons up to C₄. The second is the main reaction set comprising the same general 25 reaction classes as the *n*-heptane mechanism with rates for branched hydrocarbons up to C₈. The most recent updates to small hydrocarbon chemistry and low-temperature chemistry pathways made to the *n*-heptane mechanism in Karwat et al. [19] are not reflected in the Mehl et al. [41] *iso*-octane mechanism used in this study. Mehl et al. [41] reported satisfactory validation of their *iso*-octane mechanism with recent shock tube, rapid compression machine, and jet stirred reactor data, in addition to the original validation data used by Curran et al. [42].

2.3. *n*-Butanol mechanism

For the *n*-butanol simulations, the chemical kinetic mechanism by Black et al. [34] was used. Reactions in the Black et al. [34] mechanism consist of a sub-set of reactions for hydrocarbons up to C₄ based on the work of Donato et al. [45] and a sub-set of reactions for *n*-butanol generated using EXGAS and modified to fit the validation data. The authors reported good agreement between the modeled and experimental ignition delay times for lean and stoichiometric mixtures at all pressures investigated. Black et al. [34] found their mechanism was less accurate for rich conditions and indicated that the ratio of unimolecular decomposition to hydrogen atom abstractions may be the cause. Dilution effects (buffer gas:O₂ of 3.6:1 and 26.6:1) on ignition delay times of stoichiometric mixtures at 2.6 atm were also captured well by the authors' mechanism. Experimental species measurements by Dagaut et al. [44] made in their jet-stirred reactor for stoichiometric mixtures at 10 atm were captured reasonably well by the *n*-butanol mechanism, with the largest disagreements observed for the rate

Table 2
Initial conditions for 0-D homogeneous batch reactor simulations used in CHEMKIN. Composition is provided on a mole basis.

Fuel (–)	Fuel (%)	O ₂ (%)	Buffer gas (%)	Φ (–)	Buffer gas:O ₂ (–)	P_o (atm)	T_o (K)
<i>i</i> -C ₈ H ₁₈	1.65	20.66	77.69	1.0	3.76	9.0	600–1100 ^a
<i>i</i> -C ₈ H ₁₈	1.19	14.88	83.93	1.0	5.64	9.0	600–1100
<i>i</i> -C ₈ H ₁₈	1.19	14.88	83.93	1.0	5.64	60.0	600–1100
<i>n</i> -C ₇ H ₁₆	1.87	20.61	77.52	1.0	3.76	9.0	600–1100 ^a
<i>n</i> -C ₇ H ₁₆	1.34	14.90	83.76	1.0	5.62	9.0	600–1100
<i>n</i> -C ₇ H ₁₆	1.34	14.90	83.76	1.0	5.62	60.0	600–1100
<i>n</i> -C ₄ H ₉ OH	3.38	20.30	76.32	1.0	3.76	3.2	600–1100 ^a
<i>n</i> -C ₄ H ₉ OH	2.45	14.71	82.84	1.0	5.63	3.2	600–1100
<i>n</i> -C ₄ H ₉ OH	2.45	14.71	82.84	1.0	5.63	60.0	600–1100

^a Additional high temperature simulations (600–1800 K) were carried out for comparison to experimental data at these conditions.

of *n*-butanol consumption, ethyne (C_2H_2) concentrations, and butanal (C_3H_7CHO) concentrations.

3. Results and discussion

Pressure–time histories from the CHEMKIN simulations were used to determine ignition delay times for the *n*-heptane, *iso*-octane, and *n*-butanol mixtures. When single stage ignition occurred, the overall ignition delay time was defined as the time from the start of the simulation to the time corresponding to the maximum rate of pressure rise, dP/dt_{max} , as has been used previously [10–12,19,31]. For conditions where two stage ignition occurred, the first stage of ignition was determined from the start time of the simulation to the time of the first maxima in the rate of pressure rise. The overall ignition delay time for two stage ignition was defined as the start time of the simulation to the time of the second maxima in the rate of pressure rise. The temperatures reported for the ignition delay times and pressure–time histories were the initial temperatures of the simulations. Figure 1 compares the pressure time histories and corresponding pressure derivatives for the different buffer gases for *n*-heptane at a low temperature (700 K), low pressure (9.0 atm), and dilute mixture condition (where dilution was quantified by the molar ratio of the buffer gas to the O_2 in the mixture). As expected, the final pressures after ignition vary based on the specific heat capacity of the buffer gases. Additionally, the CO_2 results show a slow rate of pressure rise compared to the other buffer gases during ignition.

Figure 2 shows results for computed first stage ignition delay times (+ symbols) and the overall ignition delay times (solid lines) in an Arrhenius diagram for the case of *n*-heptane at $\phi = 1.0$, $P = 9.0$ atm, buffer gas: $O_2 = 5.62$, over the temperature range 600–1100 K for the four buffer gases considered. While only the results for *n*-heptane are presented in Fig. 2, similar trends were observed for *iso*-octane at the same equivalence ratio, pressure, and dilution level. (Please see the Supplemental material for simulation results not shown in the text. The Arrhenius diagram for *iso*-octane at these conditions is provided as Fig. S4.) Within the negative temperature coefficient (NTC) region (~ 650 – 850 K for *n*-heptane, ~ 600 – 800 K for *iso*-octane), buffer gas composition had significant effect on the overall ignition delay time, with argon and water vapor decreasing overall ignition times up to 31% (30% for *iso*-octane) and carbon dioxide increasing times up to 49% (65% for *iso*-octane) relative to N_2 . However, the first stage of ignition shows little variance ($<7\%$ for *n*-heptane, $<16\%$ for *iso*-octane) for any of the buffer gases relative to nitrogen. Outside of the NTC region for both fuels, the buffer gases had less impact on the computed ignition delay times, with water vapor showing the most significant effect in the intermediate

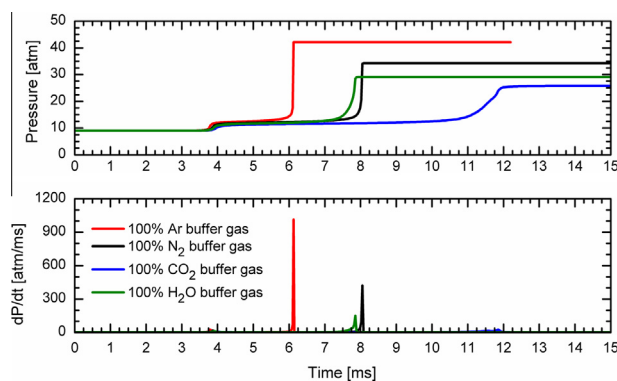


Fig. 1. Computed pressure–time histories and corresponding pressure derivatives for stoichiometric *n*-heptane mixtures. Initial conditions of 700 K, 9.0 atm, $n-C_7H_{16} = 1.34\%$, $O_2 = 14.90\%$, buffer gas = 83.76% (mole basis).

temperature chemistry regime, decreasing times up to 37% (34% for *iso*-octane). Rate of production analysis showed that OH and HO_2 radicals from water are not significant reaction pathways at the conditions studied for either PRF, and therefore are not the source of the decrease in ignition delay times at higher temperatures when water is considered as the buffer gas. Water, and carbon dioxide at higher temperatures ($T > \sim 850$ K for *n*-heptane), decrease ignition delay times by their role in three-body reactions (e.g., H_2O_2 decomposition) and the corresponding enhanced collision efficiencies of water and carbon dioxide in such reactions. In the intermediate temperature chemistry regime for *n*-heptane ($T > 900$ K), argon and carbon dioxide did not alter ignition times more than 17% (15% for *iso*-octane), which is considered reasonably within the uncertainty of the overall reaction mechanisms.

For *n*-butanol at conditions of $\phi = 1.0$, $P = 3.2$ atm, and buffer gas: $O_2 = 5.63$, there was no NTC behavior over the range of temperatures considered, and the relative reactivity due to the buffer gases was consistent throughout the temperature range. (The Arrhenius diagram for *n*-butanol at these conditions is provided as Fig. S9.) Water vapor and carbon dioxide decreased ignition times up to 52% and 32%, respectively, compared to N_2 , and argon increased ignition delay times less than 10% compared to N_2 . As with the primary reference fuels, water was not a significant source of radical formation when used as the buffer gas for the conditions studied and the decrease in ignition delay times caused by CO_2 and H_2O is explained by enhanced collision efficiencies.

Figure 3 presents the pressure time histories for the buffer gases at a high pressure condition of 60 atm and 700 K. The duration of the first stage of ignition is considerably reduced for each buffer gas compared to the same temperature and lower pressure conditions of Fig. 1. The CO_2 continues to exhibit longer times for heat release relative to the other buffer gases. This high pressure condition is near the low temperature limit for the NTC behavior predicted for *n*-heptane, as seen in Fig. 4. The simulations conducted at high pressure ($\phi = 1.0$, 60.0 atm, buffer gas: $O_2 = 5.64$) show the NTC region shifted to higher temperatures compared to the low pressure simulations (to ~ 750 – 950 K for *n*-heptane, to ~ 700 – 900 K for *iso*-octane), and the general trends as a function of buffer gas composition were the same as observed at low pressure. Buffer gas effects on the first stage ignition delay times were negligible (less than 15%) for *n*-heptane and *iso*-octane. Although only two pressures were considered in this study, the results indicate the ignition delay times for the fuels scale by $\sim P^{-1}$ on the high temperature side of the NTC region.

Simulations at the lower level of dilution (3.76 molar ratio of buffer gas to O_2) and the lower pressure (9 atm) revealed similar trends to the higher dilution results as seen in Fig. 5. The NTC region was shifted slightly (<25 K) for *n*-heptane and *iso*-octane by decreasing the dilution level from 5.62:1 to 3.76:1. Outside the NTC region, the effects of dilution were approximately proportional, e.g., the 33% decrease in dilution resulted in $\sim 35\%$ decrease in ignition delay times outside of the NTC region for *n*-heptane/nitrogen and *iso*-octane/nitrogen mixtures. Within the NTC region, the decrease in dilution led to $\sim 60\%$ decrease in ignition delay times for the *n*-heptane/nitrogen and *iso*-octane/nitrogen mixtures.

As expected, decreasing the dilution de-amplified the effects of buffer gas composition to the point where changing the buffer gas composition resulted in less than 30% change in the ignition delay time throughout the range of conditions studied for *iso*-octane and *n*-heptane. Lowering the level of dilution for *n*-butanol indicated an approximately proportional impact on ignition delay time (the 33% decrease in dilution yielded a systematic decrease in ignition delay time of $\sim 20\%$). Air levels of dilution were not considered for the high pressure case ($P = 60$ atm) for any fuel.

As seen in this study, conditions of lower pressure ($< \sim 10$ atm) and higher levels of dilution (buffer gas: $O_2 > 3.76$) amplify the

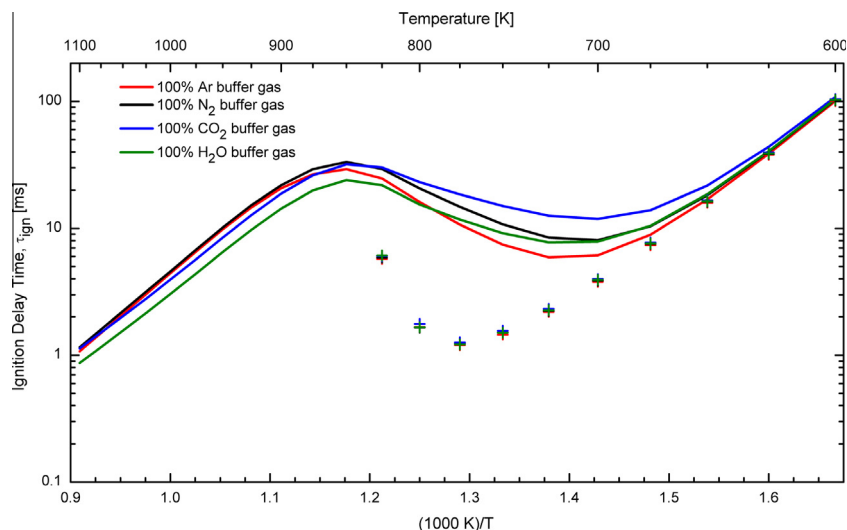


Fig. 2. Arrhenius diagram for computed ignition delay times of *n*-heptane. For conditions where two stages of ignition were observed, the + symbols indicate the computed first stage ignition delay times. Initial conditions of $P = 9.0$ atm, $n\text{-C}_7\text{H}_{16} = 1.34\%$, $\text{O}_2 = 14.90\%$, buffer gas = 83.76% (mole basis).

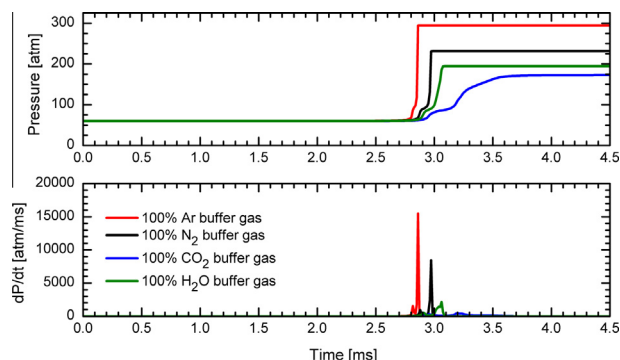


Fig. 3. Computed pressure–time histories and corresponding pressure derivatives for stoichiometric *n*-heptane mixtures. Initial conditions of 700 K, 60.0 atm, $n\text{-C}_7\text{H}_{16} = 1.34\%$, $\text{O}_2 = 14.90\%$, buffer gas = 83.76% (mole basis).

effects of buffer gas composition, and the NTC region is particularly sensitive to buffer gas composition. NTC behavior is generally attributed to the ratio of reactions forming alkylperoxy radicals

(RO_2) to reactions forming alkyl radicals (R). The subsequent isomerization of RO_2 to hydroperoxyalkyl radicals (QOOH), and resulting pathways lead to chain-branching decomposition. Increases in pressure result in a shift in favor of RO_2 formation as shown by Villano et al. [47], shifting the NTC region to higher temperatures as pressure increases. Increasing pressure also decreases the magnitude of the NTC behavior as observed in the experimental study by Ciezki and Adomeit [25]. As dilution decreases, the O_2 concentrations increase also shifting the ratio towards RO_2 formation and moving the NTC region to higher temperatures. Comparing Figs. 2, 4 and 5, pressure exhibits greater influence than dilution on the NTC behavior for the conditions studied here.

3.1. Comparison of model trends with experimental data

Simulated autoignition times for stoichiometric *n*-butanol mixtures at air levels of dilution and initial pressures of 3.2 atm are compared with existing data from the literature in Fig. 6. The experimental data shown in the figure are from stoichiometric mixtures and initial temperatures from 678 K < T < 1711 K. The

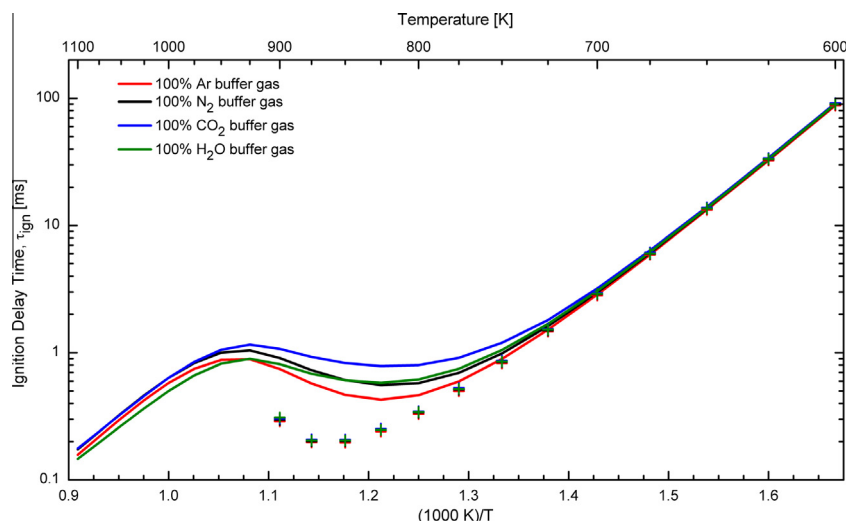


Fig. 4. Computed ignition delay times for stoichiometric *n*-heptane mixtures at high pressures. For conditions where two stages of ignition were observed, the + symbols indicate the computed first stage ignition delay times. Initial conditions of $P = 60.0$ atm, $n\text{-C}_7\text{H}_{16} = 1.34\%$, $\text{O}_2 = 14.90\%$, buffer gas = 83.76% (mole basis).

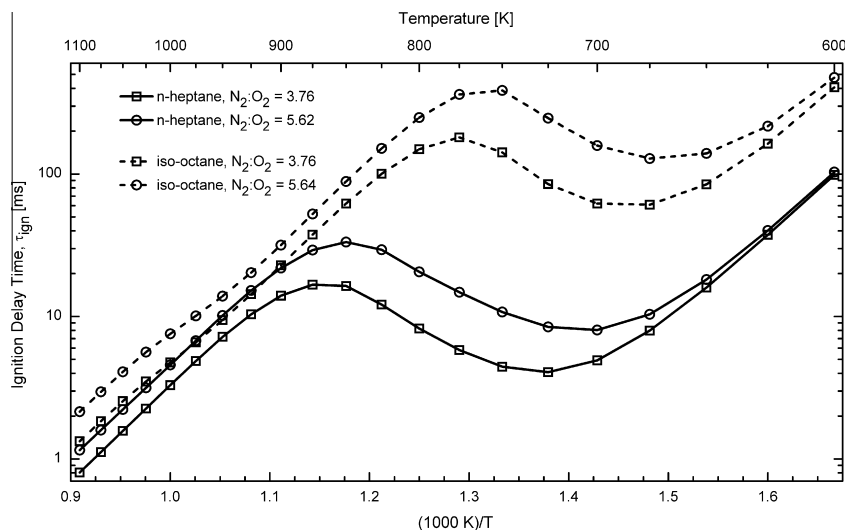


Fig. 5. Dilution level effects on ignition delay times for stoichiometric mixtures of the primary reference fuels, *iso*-octane and *n*-heptane, at low pressure ($P = 9.0$ atm) using N_2 as the buffer gas.

experimental data were obtained for a wide range of pressures and dilution levels, with pressures from 0.9 to 90.3 atm and buffer gas: O_2 ratios from 3.6 to 65.5. Based on the trends observed in this study and in previous work [19,49], all experimental ignition delay times presented in Fig. 6 were scaled assuming ignition delay times scale inversely with pressure (i.e., as P^{-1}) and proportionally with dilution based on the molar ratio of buffer gas: O_2 . All experimental data were scaled to 3.2 atm and buffer gas: $O_2 = 3.76$. These scaling rules are considered reasonable for data outside the NTC region, with higher uncertainty introduced by scaling within the NTC region.

For temperatures above ~ 1000 K, there is fair agreement amongst the experimental data, within a factor of 3, from all facilities regardless of buffer gas composition. The degree of scatter above 1000 K cannot be attributed to buffer gas effects as the majority of data were acquired with argon. For temperatures below ~ 900 K, the model predictions deviate from the experimental data. For the long ignition times (>100 ms), this may be due in part because the adiabatic 0-D model is not appropriate for many RCM

data. For example, low temperature ignition experiments may be complicated by heat losses as well as weak ignition phenomena, where heat losses typically increase experimental measurements of ignition delay times and weak ignition effects decrease ignition delay times [50–52]. Additionally, the reaction mechanisms used in the study have limited validation targets at low temperatures.

The trends between the simulations and experimental data for buffer gas composition are generally consistent. In the temperature range 927–1034 K, Karwat et al. [31] varied argon levels from 15% to 35% of the buffer gas (with the balance being N_2) for experiments at 3.2 atm and buffer gas: $O_2 = 5.63$. They observed no changes in autoignition behavior with changes to the buffer gas composition, which is consistent with the model predictions for Ar and N_2 . Further, in the limited temperature window where data from different facilities and different buffer gases overlap (~ 950 –1250 K), the results are consistent for argon and nitrogen as is predicted by the model, albeit the scatter in the experimental data is greater than the effects predicted by changing the buffer gas composition.

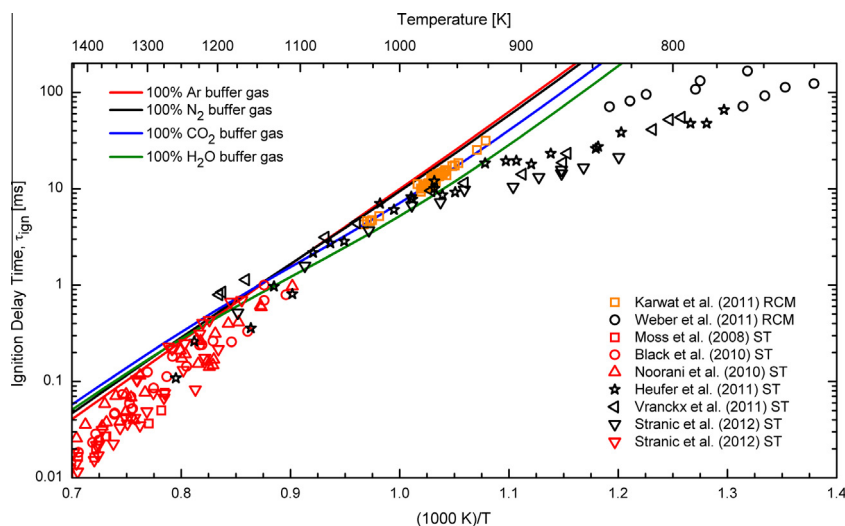


Fig. 6. Comparison of computed and experimental ignition delay times of stoichiometric mixtures of *n*-butanol at air dilution levels. Simulation results are presented as lines for initial conditions of $P = 3.2$ atm, $n\text{-C}_4\text{H}_9\text{OH} = 3.38\%$, $O_2 = 20.30\%$, buffer gas = 76.32% (mole basis). The symbols are the experimental results and are colored-coded based on the buffer gas composition used: red for argon, black for nitrogen, and orange for mixtures of Ar/ N_2 buffer gases. All experimental data have been scaled to $P = 3.2$ atm and dilution of 3.76:1 (see text for details). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Ignition delay times from the *n*-heptane simulations are compared with experimental data at air dilution levels in Fig. 7. Due to the significant NTC characteristics of *n*-heptane and the effects of pressure on NTC behavior, experiments within a narrow range of pressures (6–12 atm) were used for comparison. The dilution levels considered in the experiments were in the range of 2.48–5.62, molar ratio of buffer gas to O₂. As with the *n*-butanol comparison, the experimental data were scaled inversely with pressure and proportionally with dilution throughout the temperature range considered. The RCM studies used three buffer gases (Ar, N₂, and CO₂) and blends of the buffer gases to vary the end of compression temperatures over the range considered. There is little overlap in the experimental data using different buffer gases. However, at high temperatures ($T > 1000$ K), the shock tube data using N₂ are systematically slower than the shock tube data using Ar, with larger differences at higher temperatures. This trend extends to higher temperatures not shown on the figure. While the reaction mechanism predicts faster ignition for Ar mixtures compared to N₂, the model predicts only a slight (~5%) decrease in ignition delay time compared to the difference in the experimental data sets (a factor of ~2). Within the NTC region, the scatter in the RCM data is larger than the variation predicted by the model for the range of buffer gases considered, so no conclusions on the effect of buffer gas can be drawn.

Results of stoichiometric *iso*-octane simulations at air levels of dilution are compared with experimental data in Fig. 8. The experimental data span a pressure range of 6–12 atm, and have been scaled to 9 atm in the figure using $\tau \propto P^{-1}$. The experimental data span dilution levels of 2.48–5.00 and have been scaled to air levels of dilution using $\tau \propto (\text{buffer gas}:\text{O}_2)$. There is a small temperature range (~950–1000 K) where there are experimental data using N₂ (black symbols), Ar (red symbols), and N₂/Ar and N₂/CO₂ blends (orange symbols). The trends are consistent with the model predictions, with Ar yielding the fastest ignition and the blends yielding the slowest; however, the reaction mechanism predicts less sensitivity to the buffer gas than observed in the experiments.

3.2. Thermal effects of buffer gas composition

In the mechanisms considered in this study, typical third-body collision efficiencies for carbon dioxide and water vapor are ~2–4

and ~5–12 times that of nitrogen, respectively. The third-body collision efficiencies for argon are typically ~0.3–0.9 that of nitrogen. To isolate the thermal effects of buffer gas composition from the chemical kinetic effects of the collision efficiencies on ignition delay times, simulations were conducted where all the third-body collision efficiencies were set to 1, the collision efficiency of nitrogen. The simulations considered mixtures of $\phi = 1.0$ with buffer gas:O₂ = 5.64 and low pressures of 3.2 atm (*n*-butanol) and 9.0 atm (*n*-heptane and *iso*-octane). Results from the simulations are provided in the Supplemental material. The resulting first stage and overall ignition delay times correlated with expectations based on the specific heat capacity of the buffer gases (i.e., $\tau_{\text{Ar}} < \tau_{\text{N}_2} < \tau_{\text{H}_2\text{O}} < \tau_{\text{CO}_2}$) for all three fuels, for all temperatures.

Outside of the NTC region, buffer gases without enhanced (or reduced) collision efficiencies relative to N₂ changed the ignition delay by ~±25%. Within the NTC region for *n*-heptane (Fig. S2) and *iso*-octane (Fig. S6), the changes to the collision efficiencies changed the overall ignition delay time by nearly a factor of 2, while the first stage of ignition was virtually unchanged (<7% for *n*-heptane and <16% for *iso*-octane). The large difference in the sensitivity to buffer gas composition in the NTC region is attributed to the specific heat capacities of the buffer gases when the collision efficiencies are set to 1. Carbon dioxide and water vapor with higher specific heat capacities than nitrogen exhibit smaller pressure and temperature increases during the first stage of ignition. Since mixtures with carbon dioxide and water vapor are significantly cooler after the first stage compared to nitrogen, the second stage of ignition is demonstrably longer for H₂O and CO₂ when collision efficiencies are set equal to 1. The second stage of ignition is discernibly shorter for argon when compared to nitrogen, due to a higher second stage temperature, when the efficiencies of all the buffer gases are set to 1.

When the recommended collision efficiencies are considered, as in Figs. 2 and 7 for *n*-heptane, the effects of carbon dioxide and water vapor change from increasing ignition delay times at low temperatures (<700 K for *n*-heptane) to decreasing ignition delay times (relative to nitrogen) as temperature increases. This effect is observed at both low and high pressures and for both air and dilute mixtures for *iso*-octane and *n*-heptane, and shows how the chemical kinetic and thermal effects are particularly convolved in the NTC region.

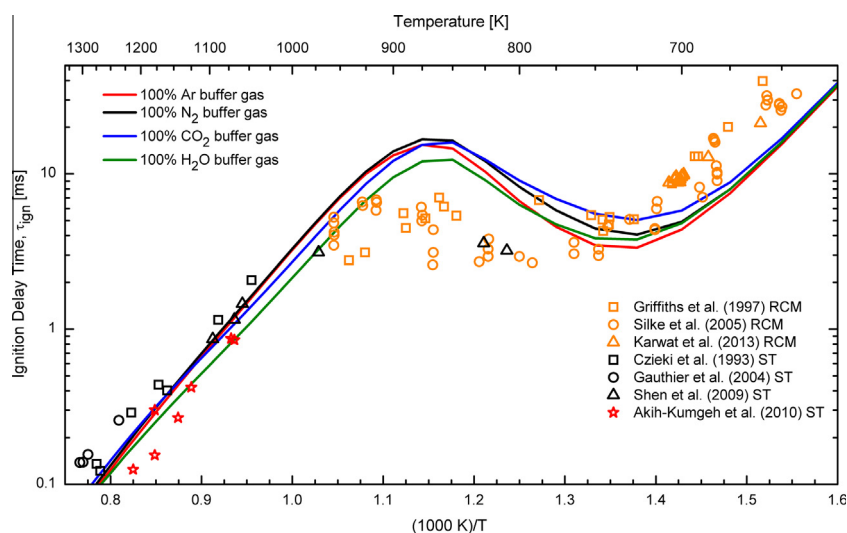


Fig. 7. Comparison of computed and experimental ignition delay times of stoichiometric mixtures of *n*-heptane at air dilution levels. Simulation results are presented as lines for initial conditions of $P = 9.0$ atm, $n\text{-C}_7\text{H}_{16} = 1.87\%$, $\text{O}_2 = 20.61\%$, buffer gas = 77.52%. Experimental results are presented as symbols and are colored-coded based on buffer gas composition: red for argon, black for nitrogen, and orange for either Ar/N₂ or CO₂/N₂ buffer gases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

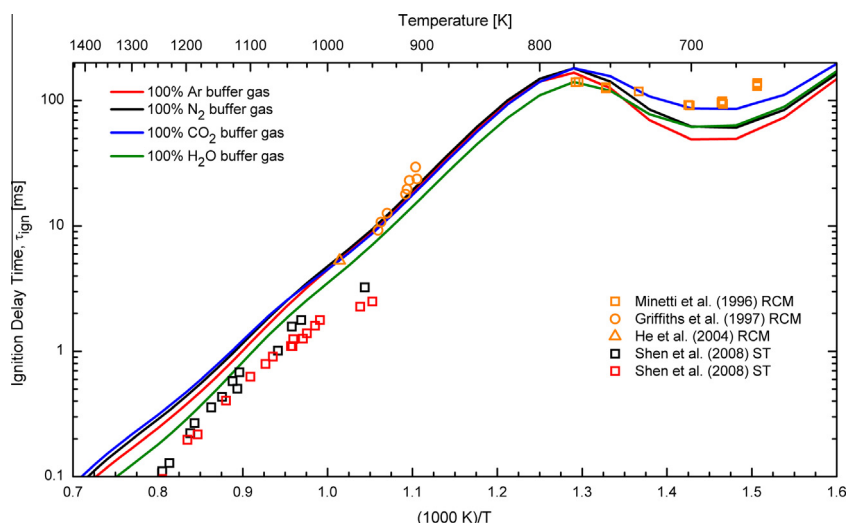


Fig. 8. Comparison of computed and experimental ignition delay times of stoichiometric mixtures of *iso*-octane at air dilution levels. Simulation results are presented as lines for initial conditions of $P = 9.0$ atm, $i\text{-C}_8\text{H}_{18} = 1.65\%$, $\text{O}_2 = 20.66\%$, buffer gas = 77.69% . Experimental results are colored based on buffer gas composition: red for argon, black for nitrogen, and orange for either Ar/N_2 or CO_2/N_2 buffer gases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3. Effects of buffer gas composition on heat release rate

As seen in Fig. 1, buffer gas composition affects heat release rates as well as ignition delay times. The maximum rate of pressure rise (MRPR) correlates with the maximum heat release rate during autoignition, and MRPR is a key metric of engine performance. A characteristic time for heat release, τ_{HR} , can be defined using the MRPR and the overall pressure rise during ignition, $P_{\text{max}} - P_0$:

$$\tau_{\text{HR}} = (P_{\text{max}} - P_0) / \text{MRPR}$$

Figure 9 shows the relationship between the simulation results for τ_{HR} and ignition delay time for *n*-heptane at $\phi = 1.0$ and the dilution levels and pressures considered in this work. Results showing the relationship for τ_{HR} and ignition delay times from *n*-butanol and *iso*-octane simulations are provided in the [Supplemental material](#). For *n*-heptane at the air levels of dilution and $P = 9$ atm, the results show characteristic times for heat release correlate linearly with the autoignition times, and the composition of the buffer gas has a larger effect on heat release at faster ignition delay times. The effects of buffer gas composition on heat release are amplified at higher levels of dilution and higher pressures, with over an order of magnitude difference between the characteristic time for heat release for CO_2 and N_2 at $P = 60$ atm for the same autoignition time. The results indicate that strategies that fix engine autoignition phasing will yield different combustion rates based on the buffer gas composition, and higher engine speeds will be affected more than lower engine speeds. However, while the impact of composition is dramatic at some conditions in Fig. 9, the concentrations of CO_2 and H_2O in EGR are generally each less than 5% mole fraction, so the effects will be reduced at expected engine levels of dilution.

3.4. Sensitivity analysis

Sensitivity of the ignition delay time and heat release rate results to the third-body collision efficiencies, η , were examined for the three fuels at the stoichiometric, lower pressures of 3.2 atm (*n*-butanol) and 9.0 atm (*n*-heptane and *iso*-octane), and higher level of dilution (buffer gas: $\text{O}_2 = 5.64$) conditions. Third-body collision efficiencies were varied by a factor of 2 from their nominal values, η_0 , for this local sensitivity analysis. The factor of 2 was

considered a reasonable estimate of the uncertainties of the collision efficiencies. The authors of the *n*-heptane mechanism used a separate reaction, (R20), to represent hydrogen peroxide decomposition with water as a third-body reactant rather than include a third-body collision efficiency for water in (R18). Consequently, the sensitivity results for water were computed by varying the low-pressure A-factor of (R20) by a factor of 2. The local sensitivity coefficient was defined as the difference between the new and the baseline ignition delay times (i.e., ignition delay times with modified collision efficiencies and the nominal values respectively), normalized by the baseline ignition delay time. Two reactions were found to have significant impact on the predicted ignition delay times and heat release rates:



All other reactions involving 3rd body collision efficiencies had negligible impact on the calculated ignition delay times.

Results of the sensitivity analysis for ignition delay time are presented in Fig. 10 for *n*-heptane at the conditions of $\phi = 1.0$, $P = 9.0$ atm, buffer gas: $\text{O}_2 = 5.62$. The results show a factor of 2 uncertainty in the values of the collision efficiencies can have as much as a 23% effect on ignition delay time for temperatures ≥ 750 K. For all fuels, the sensitivities for all the buffer gases remained below 30% for (R18) throughout the entire temperature range at this pressure and dilution level. Sensitivity analysis of (R9) revealed that ignition delays were affected less than 10%/30%/5% for *n*-heptane/*iso*-octane/*n*-butanol respectively at this same initial condition.

Results of the sensitivity analysis for representative pressure and pressure derivative time histories are shown in Fig. 11 for $T = 700$ K for *n*-heptane at $\phi = 1.0$, $P = 9.0$ atm, buffer gas: $\text{O}_2 = 5.62$ and using CO_2 as the buffer gas. The first stage of ignition was unaffected by changes in the collision efficiencies for all fuels and buffer gases. Varying the collision efficiency of (R9) resulted in less than a 5% change of the ignition delay time for the *n*-heptane/carbon dioxide mixture. However, (R9) significantly altered the maximum rate of pressure rise in the second stage of ignition for the *n*-heptane/carbon dioxide mixture, by nearly a factor of two between the two limiting values of the collision efficiencies. The results

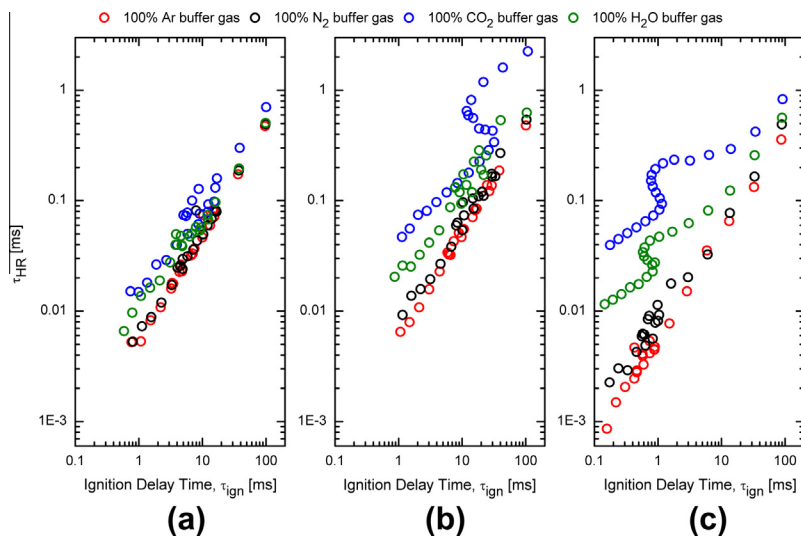


Fig. 9. Simulation results for characteristic times of heat release as a function of ignition delay time for stoichiometric *n*-heptane mixtures at initial conditions of: (a) $P = 9.0$ atm, buffer gas: $O_2 = 3.76$, (b) $P = 9.0$ atm, buffer gas: $O_2 = 5.62$, (c) $P = 60.0$ atm, buffer gas: $O_2 = 5.62$.

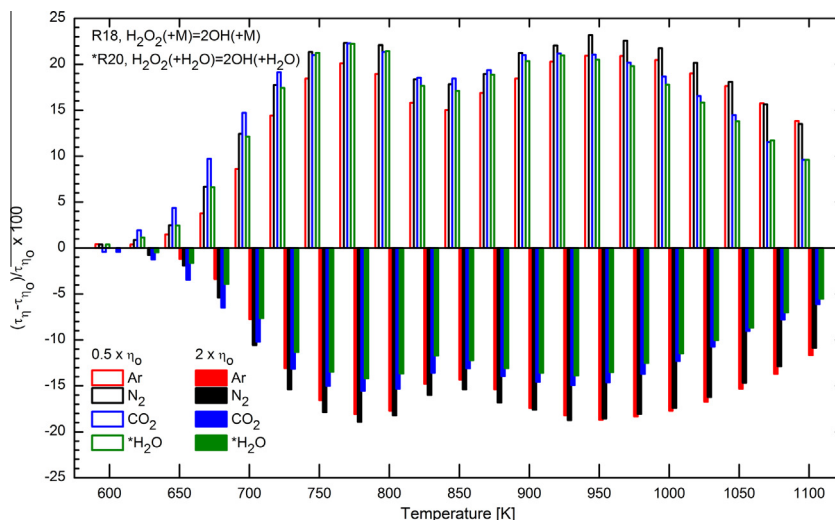


Fig. 10. Results of the sensitivity analysis of ignition delay time for *n*-heptane to changing the third body collision efficiencies of the reaction (R18) $H_2O_2 (+M) = 2OH (+M)$ by a factor of 2. The initial conditions of the simulations were $P = 9$ atm, $n-C_7H_{16} = 1.34\%$, $O_2 = 14.90\%$, buffer gas = 83.76%. *The H_2O sensitivity coefficients were determined by changing the A factor for reaction (R20) by a factor of 2. See text for details.

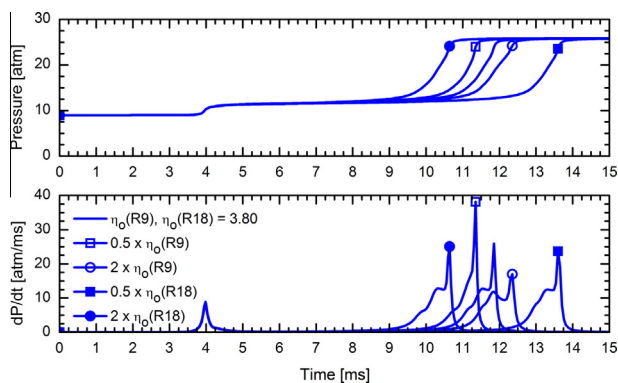


Fig. 11. Results of changing the collision efficiencies of reactions (R9) and (R18) on pressure and pressure derivative time histories of *n*-heptane at initial conditions of $T = 700$ K, $P = 9.0$ atm, $n-C_7H_{16} = 1.34\%$, $O_2 = 14.90\%$, $CO_2 = 83.76\%$.

for (R18) show while the ignition delay time was affected significantly, the maximum rate of pressure rise was nominally unchanged (<9%) for *n*-heptane/carbon dioxide. Water vapor showed similar effects to carbon dioxide on the pressure rise rate for *n*-heptane, while argon and nitrogen exhibited little sensitivity (<14% change). The pressure derivative time histories were less affected (<20% change) by changes in the collision efficiencies for (R9) and (R18) at 700 K for *iso*-octane or *n*-butanol for any of the buffer gases.

4. Conclusions

This study improves our understanding of the thermal and chemical kinetic effects of buffer gas composition on the important combustion characteristics of ignition and heat release rate for three reference fuels. Simulations were presented for *n*-heptane, *iso*-octane, and *n*-butanol in stoichiometric mixtures with four buffer gases (Ar, N_2 , H_2O , and CO_2) for a range of pressures, dilution levels, and temperatures. Based on the conditions examined in this

study, low pressures (<10 atm), high levels of dilution ($>1.5\times$ that of air), and negative temperature coefficient conditions are most likely to result in significant thermal and chemical kinetic effects on ignition delay time and heat release rates due to the composition of buffer gases. Fuels that exhibit negative temperature coefficient behavior may show significantly more pronounced buffer gas effects during two stage ignition, with a factor of 2 or more impact on overall ignition times. Caution should be exercised when comparing and compiling data acquired in the NTC region from different experimental facilities that may have used different buffer gases. Ignition data are also subject to uncertainties in the collision efficiencies of the different buffer gases. Experimental data for *n*-heptane at high temperatures indicated a faster ignition for Ar compared to N_2 (on the order of a factor of 2); however, the model simulations predicted much smaller sensitivity to buffer gas composition for *n*-heptane. There are currently few experimental studies which target the effects of buffer gas composition on ignition; particularly studies that vary buffer gas composition using the same facility. Additional experimental and computational studies will reduce the chemical kinetic uncertainties associated with buffer gas composition, and such work has potential to significantly improve predictive understanding of the complex chemistry and thermal interactions of NTC behavior.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.09.022>.

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